



Oxygen-defective MnO_{2-x} rattle-type microspheres mediated singlet oxygen oxidation of organics by peroxymonosulfate activation

Jie Yu^{a,b}, Tao Zeng^{c,*}, He Wang^a, Hui Zhang^a, Yanping Sun^a, Lei Chen^a, Shuang Song^c, Lingxiangyu Li^d, Huixiang Shi^{a,*}

^a College of Environment & Resource Sciences, Zhejiang University, Hangzhou 310058, PR China

^b College of Quality & Safety Engineering, Department of Environment Engineering, China Jiliang University, Hangzhou 310018, PR China

^c College of Environment, Zhejiang University of Technology, Hangzhou 310032, PR China

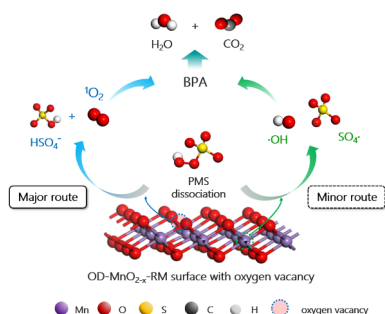
^d Department of Chemistry, School of Sciences, Zhejiang Sci-Tech University, Hangzhou 310018, PR China

HIGHLIGHTS

- Defect engineering and morphology controlling were utilized to design OD- MnO_{2-x} -RM.
- Rattle-type morphology enhances the mass transfer and accessibility of active sites.
- Oxygen vacancies enable strong affinity for PMS and promote interfacial charge transfer.
- $^1\text{O}_2$ rather than $\text{HO}^\cdot$ and $\text{SO}_4^{\cdot-}$ was verified as the dominant ROS for BPA degradation.
- OD- MnO_{2-x} -RM/PMS is one of the best-performing Mn-based systems ever reported for BPA removal.

GRAPHICAL ABSTRACT

Oxygen-defective MnO_{2-x} rattle-type microspheres were synthesized as robust and enhanced heterogeneous catalyst for peroxymonosulfate activation.



ARTICLE INFO

Keywords:

Manganese oxides
Rattle-type structure
Peroxymonosulfate
Oxygen vacancies
Singlet oxygen

ABSTRACT

Cost-effective and low-toxicity manganese oxides (MnO_x) have been extensively used for environmental remediation. However, their unsatisfactory activity and intrinsically poor electronic conductivity hamper the application and leave considerable room for improvement. Utilizing defect engineering and morphology controlling, we herein designed MnO_{2-x} rattle-type microspheres with plenty of oxygen-defective MnO_2 nanoflakes vertically aligned on the surface (OD- MnO_{2-x} -RM) as a novel peroxymonosulfate (PMS) activator for bisphenol A (BPA) removal. Benefiting from the proper presence of oxygen vacancies (OVs) and unique nanoarchitecture, the OD- MnO_{2-x} -RM (40 min) outperformed all other benchmark catalysts [such as $\text{MnO}_2/\text{MnCO}_3$, MnO_2 -RM, OD- MnO_{2-x} -RM (20, 60, and 80 min)]. The turnover frequency (TOF) of optimized OD- MnO_{2-x} -RM catalyst reaches up to 83.3 min^{-1} , making it one of the best-performing Mn-based catalysts ever reported for BPA removal via PMS activation. More interestingly, unlike other reports, $^1\text{O}_2$, instead of $\text{OH}^\cdot$ and $\text{SO}_4^{\cdot-}$, was found to be the main reactive specie responsible for BPA oxidation. Various experimental and density functional theory (DFT) findings indicate that these surface oxygen-deficient sites with abundant localized electrons not only accelerate the interfacial electronic conduction and charge rearrangement but also favor chemisorption of PMS, however OVs in excess are superfluous and even counteractive for reaction kinetics. This structure arrangement simultaneously optimizes the active site exposure, changes the active species binding, transfer, and activation, which may pave a new path for rationally fabricating transition metal oxide-based catalysts for water treatment.

* Corresponding authors.